When Does Synthetic Data Help?

A Survey of Quality, Diversity, and Utility in Text Classification with Limited Labeled Data

Team 15 | Five-member team | 12 October 2026

1 Background and Motivation

Generating text can ease label scarcity, but fluency and variety do not guarantee useful supervision. Building on a broader augmentation survey [5], we will synthesize when synthetic data improves classification, considering label correctness, distribution mismatch, and supervision requirements.

2 Scope and Research Questions

The scope is generative augmentation for **sentiment, topic, and intent classification**. Zero-shot, few-shot-per-class, and larger low-label subsets are separate settings. Only in-scope experiments from CoDa, ABEX, DoAug, and LLM2LLM enter core evidence; NER, QA, and GSM8K results are excluded. Other classification tasks (e.g., entailment, acceptability, and relation classification) are contextual.

- **RQ1:** Under what supervision budgets and task conditions do generation strategies improve classification over their baselines?
- **RQ2:** When does diversity support useful coverage or introduce label noise, according to controlled comparisons?
- **RQ3:** When do filtering, correction, and weighting improve performance? Mixing ratios, stability, and costs are secondary, discussed only where evaluated; no universal optimum is assumed.

3 Review Method and Preliminary Taxonomy

Search. In Weeks 1–2, we will search ACL Anthology, ICLR proceedings/OpenReview, ICML/PMLR, and NeurIPS proceedings: primarily 2023–2026 papers available by **11 October 2026**, plus earlier background. Logs will record actual dates, sources, exact queries, and results. Query: ("synthetic data" OR "data augmentation") AND ("text classification" OR "sentiment" OR "topic classification" OR "intent"); variants add "few-shot," "limited labels," "negative results," "no improvement," or "failure." Syntax will follow each index; citation searches supplement retrieval.

Screening. Deduplication, abstract screening, and full-text review will yield at least **20 references**, including background, then **6–8 core studies**: in-scope tasks, generated training text, stated budget, and baseline. Core selection prioritizes inspectable comparisons and mechanism/budget coverage, irrespective of gain direction. Null/negative results remain eligible. A second member checks selection and exclusion reasons (duplicates, task mismatch, missing budget/baseline). Surveys are background evidence only.

Definitions. Quality separates *label fidelity* (semantic correctness) from *distribution affinity* (similarity to real data). Diversity separates lexical variation from semantic spread/coverage. Utility is accuracy/macro-F1 change on held-out real data. We will record metric definitions and human/model evaluators. Embedding similarity and automated relabeling are proxies, not proof of correct labels; numerical comparisons require matched measures and protocols.

Taxonomy. The three axes overlap; extra generator training and resources are explicit:

Axis	Initial methods and mechanisms
Generation	CoDa [1]: natural-language prompt constraints, not hard decoding constraints. ABEX [2]: LLM-built abstract–document pairs train BART; AMR editing supplies abstractions for expansion, requiring AMR tools and upstream data. DoAug [3]: paraphraser SFT and diversity-oriented DPO.
Curation	PromptMix [7]: relabeling; DDAIR [8]: ambiguity detection and regeneration/drop; DoAug [3]: informative coreset selection and diversity-based candidate selection.
Utilization	Sample weighting [4]; LLM2LLM [6]: learner-feedback generation and iterative training, also spanning the generation axis.

Evidence extraction. Each row identifies the paper, dataset/split, setting, and source table/section:

- **Supervision:** real training labels (total/per class); prompt exemplars and human-written class information; auxiliary labels for quality estimation, weighting, or discriminator training; validation size and uses for tuning/model selection; test isolation and checkpoint-selection rule. Uses and overlap will be recorded separately, not collapsed into an equivalent label count.
- **Method/resources:** generator and classifier versions; SFT/DPO or other generator supervision; external data/tools, human review, synthetic volume, mixing ratio, curation, and weighting. Generation, selection, and training costs retain their reported units and hardware context.
- **Results:** baseline and ablation, dataset-specific metric, score/change, seed count, SD/CI, significance if tested, and gains, null results, regressions, or failure conditions. Missing fields are *not reported*; absent variance is not evidence of stability.

Protocol caveats. PromptMix uses full validation data for tuning despite few-shot training (Sec. 5.3) [7]. LLM2LLM selects checkpoints using TREC test performance (App. A.3) [6]; affected rows will be flagged for test-selection bias and excluded from clean-test performance synthesis, while retaining mechanism analysis. Auxiliary real data used for weighting [4] will be recorded separately from classifier training labels.

Evidence strength. (1) Within-paper controlled ablations support mechanism effects only in their tested settings. (2) Same-task comparisons with differing protocols suggest trends with explicit confounders. (3) Different tasks/architectures support mechanism comparison, not score ranking. (4) Authors' explanations remain hypotheses. No causal claim will follow from cross-paper score differences. For example, DDAIR's 5-shot results favor dropping ambiguous samples on MPGT but iterative regeneration on BANKING77/CLINC150 (Sec. 4.3, Table 1) [8]; filtering is therefore a conditional finding, not a general guarantee.

4 Expected Outcomes and Feasibility

Deliverables: a traceable evidence table and screening log; a taxonomy comparing strengths, limits, and useful conditions; an **8-page English survey** in the official ACL template (references/appendices excluded), with at least 20 references, open questions, slides, and individual contributions. Synthesis requires no GPU; any illustrative experiment is optional.

5 Team Members and Responsibilities

Member	Primary responsibility	Specific deliverables
Hao JIANG	Scope and coordination	Search/screening log; coverage and selection-bias analysis; integration.
KeYan ZHA	Generation strategies	Mechanisms, generator training, and resource requirements; RQ1.
BoHan YANG	Quality and diversity	Definitions, proxy validity, ambiguity, and coverage evidence; RQ2.
XiangYou YU	Utilization and evaluation	Supervision/protocol audit; filtering and weighting comparisons; RQ3.
Jian TANG	Cross-paper synthesis	Evidence strength, conflicts, and failure conditions; tables and slides.

Each member will read, extract, and write substantive analysis; another member will verify core evidence and major claims. All will present, with contributions recorded.

6 Project Timeline

Stage	Tasks and deliverables
By 12 October 2026	Submit the proposal; confirm scope, initial literature, and responsibilities.
Weeks 1–2	Log searches and exclusions; collect at least 20 papers; select 6–8 core studies; extract evidence.
Weeks 3–4	Audit budgets/protocols; compare core evidence; deliver milestone taxonomy, report, and slides.
Final stage	Complete survey and claim checks, final slides, contributions, and presentation rehearsal.

Weeks are counted from proposal submission. Milestone and final deadlines will follow the instructor's announcements.

Preliminary References

[1] Chandra Kiran Evuru et al. 2024. CoDa: Constrained Generation based Data Augmentation for Low-Resource NLP. *Findings of NAACL*.

[2] Sreyan Ghosh et al. 2024. ABEX: Data Augmentation for Low-Resource NLU via Expanding Abstract Descriptions. *ACL*.

[3] Zaitian Wang et al. 2025. Diversity-oriented Data Augmentation with Large Language Models. *ACL*.

[4] Hsun-Yu Kuo et al. 2025. Not All LLM-Generated Data Are Equal: Rethinking Data Weighting in Text Classification. *ICLR*.

[5] Bosheng Ding et al. 2024. Data Augmentation using LLMs: Data Perspectives, Learning Paradigms and Challenges. *Findings of ACL*.

[6] Nicholas Lee et al. 2024. LLM2LLM: Boosting LLMs with Novel Iterative Data Enhancement. *Findings of ACL*.

[7] Gaurav Sahu et al. 2023. PromptMix: A Class Boundary Augmentation Method for Large Language Model Distillation. *EMNLP*.

[8] Galo Castillo-López et al. 2026. How DDAIR you? Disambiguated Data Augmentation for Intent Recognition. *EACL (Short Papers)*.